PRACTICAL NUMBER: 2

AIM: Implementing one data structure using another data structure: Queue using Stack.

***PROGRAM*:**

```
//sheikh wasimuddin 60 cseiot 23071044//
#include <stdio.h>
int stack1[100];
int stack2[100];
int top1 = -1;
int top2 = -1;
void push1(int value) {
   if (top1 < 100 - 1) {
      stack1[++top1] = value;
   } else {
      printf("Stack1 Overflow\n");
   }
}
void push2(int value) {
   if (top2 < 100 - 1) {
      stack2[++top2] = value;
   } else {
      printf("Stack2 Overflow\n");
   }
```

```
}
int pop1() {
    if (top1 >= 0) {
        return stack1[top1--];
    } else {
        printf("Stack1 Underflow\n");
        return -1;
    }
}
int pop2() {
    if (top2 >= 0) {
        return stack2[top2--];
    } else {
        printf("Stack2 Underflow\n");
        return -1;
    }
}
void enqueue(int value) {
    push1(value);
    printf("%d enqueued into the queue.\n", value);
}
int dequeue() {
    if (top2 == -1) {
        while (top1 >= 0) {
```

```
            push2(pop1());
        }
    }
    if (top2 >= 0) {
        int value = pop2();
        printf("%d dequeued from the queue.\n", value);
        return value;
    } else {
        printf("Queue is empty\n");
        return -1;
    }
}
void display() {
    int i;
    if (top1 == -1 && top2 == -1) {
        printf("Queue is empty\n");
    } else {
        printf("Queue elements: ");
        for (i = 0; i <= top2; i++) {
            printf("%d ", stack2[i]);
        }
        for (i = top1; i >= 0; i--) {
            printf("%d ", stack1[i]);
        }
```

```
        printf("\n");}
}
int main() {
    int choice, value;
    do {
        printf("\nQueue using Stack Menu:\n");
        printf("1. Enqueue\n");
        printf("2. Dequeue\n");
        printf("3. Display\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                printf("Enter the value to enqueue: ");
                scanf("%d", &value);
                enqueue(value);
                break;
            case 2:
                dequeue();
                break;
            case 3:
                display();
                break;
```

```
        case 4:
            printf("Exiting...\n");
            break;
        default:
            printf("Invalid choice, please try again.\n");
    }}while (choice != 4);
  return 0;
}
```

OUTPUT:

main.c

```c
//sheikh wasimuddin 60 cseiot 23071044

#include <stdio.h>
int stack1[100];
int stack2[100];
int top1 = -1;
int top2 = -1;


void push1(int value) {
    if (top1 < 100 - 1) {
        stack1[++top1] = value;
    } else {
        printf("Stack1 Overflow\n");
    }
}


void push2(int value) {
    if (top2 < 100 - 1) {
        stack2[++top2] = value;
    } else {
        printf("Stack2 Overflow\n");
    }
}


int pop1() {
    if (top1 >= 0) {
        return stack1[top1--];
    } else {
        printf("Stack1 Underflow\n");
        return -1;
    }
}
```

```
Queue using Stack Menu:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter the value to enqueue: 33
33 enqueued into the queue.

Queue using Stack Menu:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter the value to enqueue: 22
22 enqueued into the queue.

Queue using Stack Menu:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter the value to enqueue: 11
11 enqueued into the queue.

Queue using Stack Menu:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 3
Queue elements: 11 22 33

Queue using Stack Menu:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 2
```

main.c

```c
//sheikh wasimuddin 60 cseiot 23071044

#include <stdio.h>
int stack1[100];
int stack2[100];
int top1 = -1;
int top2 = -1;


void push1(int value) {
    if (top1 < 100 - 1) {
        stack1[++top1] = value;
    } else {
        printf("Stack1 Overflow\n");
    }
}


void push2(int value) {
    if (top2 < 100 - 1) {
        stack2[++top2] = value;
    } else {
        printf("Stack2 Overflow\n");
    }
}


int pop1() {
    if (top1 >= 0) {
        return stack1[top1--];
    } else {
        printf("Stack1 Underflow\n");
        return -1;
    }
}
```

```
2. Dequeue
3. Display
4. Exit
Enter your choice: 1
Enter the value to enqueue: 11
11 enqueued into the queue.

Queue using Stack Menu:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 3
Queue elements: 11 22 33

Queue using Stack Menu:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 2
33 dequeued from the queue.

Queue using Stack Menu:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 3
Queue elements: 11 22

Queue using Stack Menu:
1. Enqueue
2. Dequeue
3. Display
4. Exit
Enter your choice: 4
Exiting...


...Program finished with exit code 0
Press ENTER to exit console.
```

CONCLUSION: This practical was performed successfully.